

What Modelica is and what you can do with it

Modelica is an object-oriented, declarative, multi-domain modelling language for component-oriented modelling of complex systems, like systems containing mechanical, electrical, electronic, hydraulic, thermal, control, electric power or process-oriented sub-components.

On the face of it, Modelica seems similar to C++, Java or even Verilog. What differentiates it from other object-oriented programming languages is that it is a modelling language that mainly uses mathematical equations, which are translated into algorithmic code in another language (typically C) and then compiled and executed together with a simulation engine. It looks at blocks of algorithmic code as equation blocks that have no pre-defined causality. However, algorithmic statements inside Modelica function bodies have the usual algorithmic causality.

There are many simulation environments for Modelica. Some commercial ones are Converge, Dymola, MapleSim, SimulationX and Wolfram SystemModeler, whereas Modelicac, Jmodelica.org and OM are free simulation environments. Of the free ones, Modelicac only supports a small sub-set of Modelica, Jmodelica.org does not include a graphic model editor in its free version and only OM offers a complete free environment.

tation, variables that can be used in your code and messages displayed, especially the ones citing errors.

Also, Welcome Perspective shows a list of recent files and latest news to help you create better models; you create your model in Modelling Perspective and you can view the results of simulation in Plotting Perspective tabs.

Edit your graphical models with ease

OM graphic connection editor called OMEdit makes the process of creating, browsing, editing and simulating models smooth. Usage of pre-defined models from standard libraries for creating your own user-defined models for use and component interfaces to interconnect components makes the design process simple.

Further, sub-systems for specifying simulation parameters like time, values and more, and a plotting interface to create plots from simulated models add glitter.

Generate outputs in varied formats

OM supports generation of output files in formats like mat, plt, csv and empty. The kind of output you require can be decided based on the type of data required, speed of simulation and size of the file. When using external scripts or tools like GNU

plot to generate plots or process data, csv and plt formats come handy.

Generating outputs in mat format will allow you to post-process the same in MATLAB or OCTAVE. You can even specify the variables that should be passed onto the output file.

Using commands, you can also obtain a mathematical representation of a model using an Extensible Markup Language (XML) representation or a MATLAB representation.

As always, graphs can be gener-

ated for quick and clear analysis, and these can be time or parametric plots. There is also a 3D visualisation library called Modelica3D.

Make the most of your Python skills

Calling external functions from C comes automatically with OM. Calling external Python Code from the same, and vice versa, is an added advantage. OM also provides a scripting interface for Python, called OMPython, and PySimulation, a Python package that provides simulation and post processing/analysis.

Analyse, debug and optimise

If the model being simulated has non-linear equation systems, it is useful to enable performance analysis while simulating, as the tool would then take into account execution times of user-defined functions as well as linear, non-linear and mixed-equation systems. A profiling test also comes in handy for analysis.

Fault-tolerance levels can be studied using Figaro extensions, to get a grip of static or dynamic reliability analysis. Debugging involves going through the transformations made and the algorithms processed.

To optimise the model under scrutiny, OM offers a few facilities. Built-in dynamic optimisation is provided using OM and IpOpt. CasADi framework can also be used for the same. For the case of a static optimisation, the classical Parameter Sweep Optimisation using OMOptim is recommended.

OM finds its way to IIT Bombay

As mathematical modelling and simulation is indispensable for progress in all engineering areas, it is no surprise that OM has found users at IIT Bombay. Prof. Kannan Moudgalya from the institution says that the open source electronics design automation software eSim,

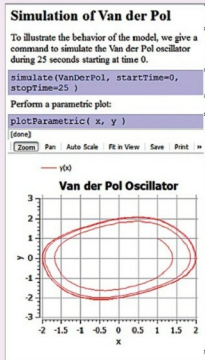


Fig. 2: Parametric plot of Van der Pol simulation